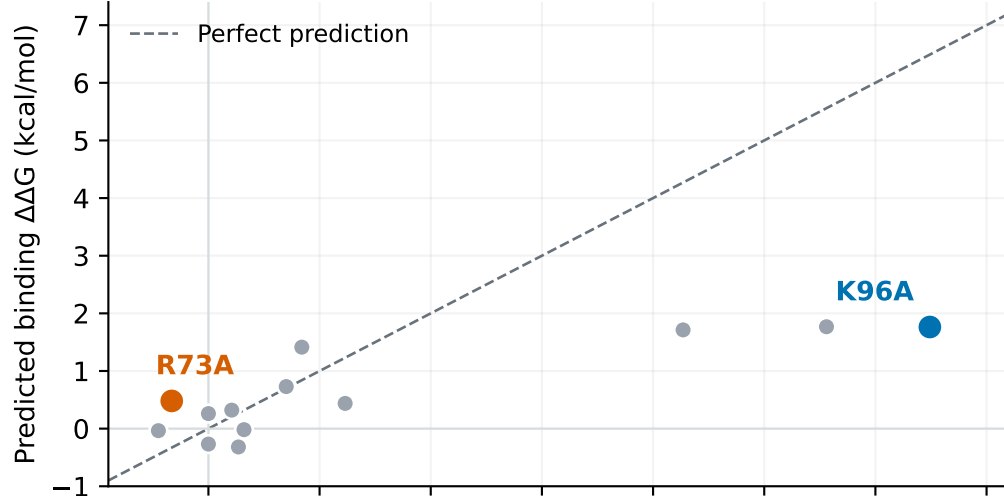
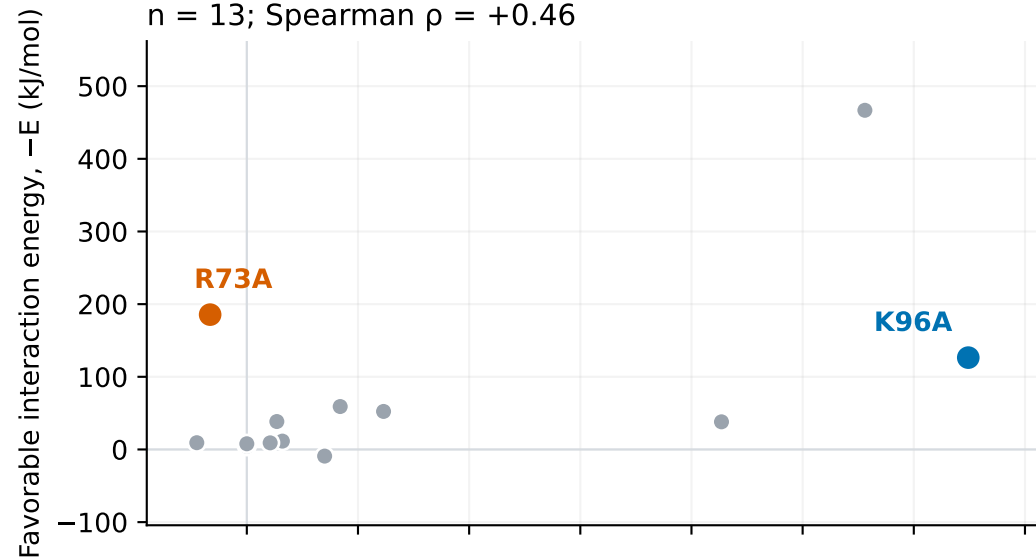


3HFM / HyHEL-10-lysozyme: learned $\Delta\Delta G$ versus simulation signals

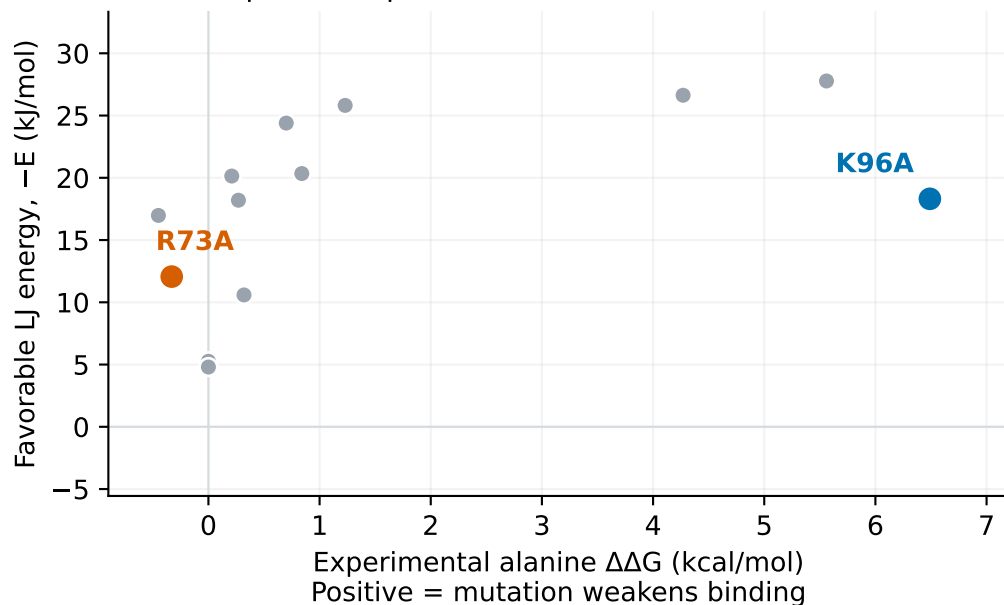
StaB-ddG · 13 single mutants
n = 13; Spearman ρ = +0.75



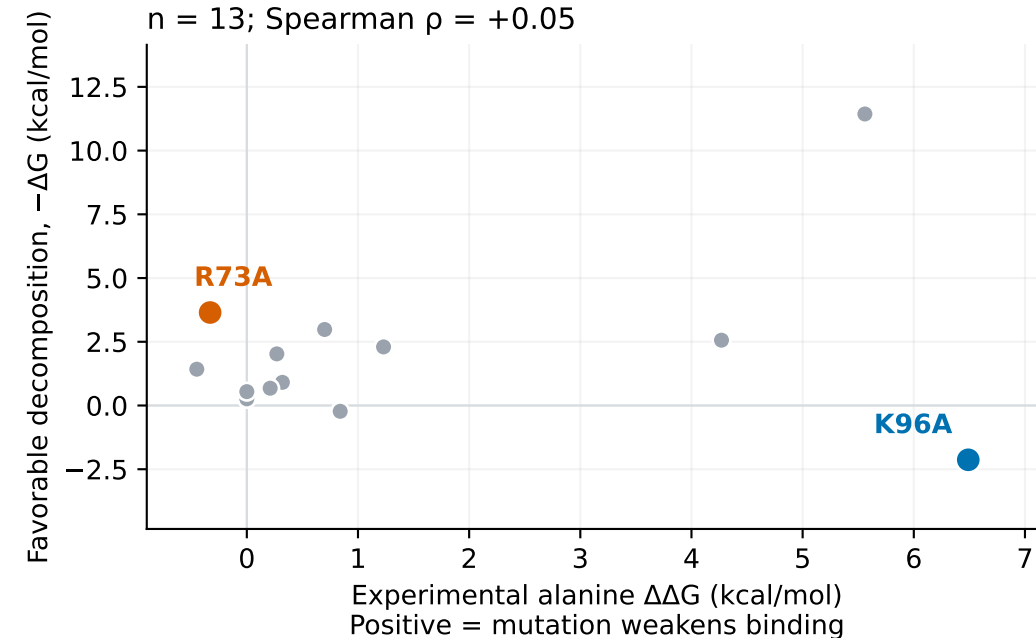
Raw interaction energy · 201 frames
n = 13; Spearman ρ = +0.46



Lennard-Jones only · 201 frames
n = 13; Spearman ρ = +0.73



MM-GBSA · 20 frames
n = 13; Spearman ρ = +0.05



Blue: K96A Orange: R73A Gray: other measured antigen residues
Simulation panels show sign-flipped wild-type energy contributions, not predicted mutation $\Delta\Delta G$.
13 mutations have ML predictions; training overlap has not been established.